Suraj Adsul

// Senior Software Engineer — AI Agent Systems

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› Summary

Senior Software Engineer with 11 years of backend and full-stack experience, now focused on shipping production AI agent systems. Currently leading AI engineering on SmartDealer-AI — a LangGraph multi-agent platform with MCP tooling and Langfuse observability — serving automotive dealer analytics. Track record of architecting B2B/B2C products end-to-end, mentoring teams, and turning research-grade LLM patterns into reliable software.

› Skills

AI / ML	Claude API · OpenAI API · LangChain · LangGraph · MCP (Model Context Protocol) · RAG · Agent Systems · Prompt Engineering · Embeddings · Vector Search · Langfuse · whisper.cpp
Languages	TypeScript · JavaScript · Python · Go · PHP · Swift · SQL · Shell
Backend & Data	Node.js · Laravel · Symfony · Express · GraphQL · REST · PostgreSQL · MySQL · NoSQL · MongoDB
Frontend	React · Next.js · Tailwind CSS · ElectronJS
Infra & DevOps	AWS (Lambda, Serverless) · Docker · CI/CD · Git · TDD · PHPUnit · Playwright · Puppeteer

› Experience

Senior Software Engineer — Tech9 (Cogitech Solutions Pvt. Ltd.) Jun 2019 – Present

Pune, India

Lead AI engineering on **SmartDealer-AI**, a multi-agent analytics platform for automotive dealers: designed domain agents (financial, game day, inventory) beneath a **LangGraph orchestrator**, wrapping 88 raw-data endpoints with LangChain-skill business logic (gross profit, fiscal skew, field transformations).

Built an **MCP (Model Context Protocol) server layer** exposing analytics tools to Claude Code and Deep Agents CLI; integrated **Langfuse** tracing for agent observability, prompt versioning, and evaluation workflows.

Implemented **robust memory and retrieval mechanisms** — conversational memory, session state, and **RAG over dealer-specific context** (reports, schemas, taxonomies) via embeddings and vector search — grounding agent responses across multi-turn analytical conversations.

Replaced pre-aggregated report endpoints with raw-data + AI calculation pipelines; replicated legacy reports (Income Statement, Make/Model Turnover, Game Day) as composable, auditable skills returning verifiable numbers instead of black-box summaries.

Authored prompt-engineering standards, guardrails, and retry/fallback policies for production LLM calls across Claude and OpenAI APIs; cut hallucinated calculations to near zero by grounding every response in deterministic Python calculators.

Led a cross-functional team of **5 engineers** shipping B2B/B2C e-commerce and legal-tech products on **PHP, Go, React, REST/GraphQL, AWS Lambda, MySQL**; owned architecture reviews, sprint planning, and hiring.

Introduced **Test-Driven Development** across the team, reducing reported production bugs by **50%** and lifting deploy confidence.

Architected a **custom Stripe-based financing solution** for an e-commerce platform, driving a **40% revenue increase** and eliminating per-transaction third-party fees.

Led migration of a legacy **Magento** storefront to a custom **Laravel** application — unlocking scalable custom-product configurators and removing recurring licensing costs.

Streamlined delivery via **CI/CD pipelines**, rigorous code review, and layered load / stress / unit / E2E testing (**PHPUnit, Playwright, Puppeteer**).

Software Engineer — RB group.aero `Aug 2018 - Jun 2019`

Pune, India

Built aviation crew-scheduling and roster-management systems (**PHP, Laravel, React, AWS, MySQL**) supporting **50+ airlines**, **99.9% uptime**, and **50,000+ end users** across web and mobile.

Designed and shipped **REST APIs and microservices** delivering scheduling, disruption, and roster data to pilot- and crew-facing clients in near real time.

Extended the roster-parser engine to **20+ new airlines** (PDF, CSV, proprietary formats), cutting end-to-end processing time by **25%**.

Hardened the parsing pipeline with integration tests that raised defect detection by **50%**, preventing silent schedule-ingestion errors from reaching operations teams.

Integrated **PCI-compliant payment gateways** for secure multi-currency transactional flows.

Collaborated directly with airline operations and product stakeholders to translate regulatory scheduling rules into working software.

Software Developer — Aloha Technology Pvt. Ltd. `Sep 2015 - May 2018`

Pune, India

Built and scaled a digital advertising platform generating **$2M in annual revenue** (**PHP, Laravel, jQuery, MySQL** on MVC).

Designed and maintained **REST APIs** serving **100,000+ monthly active users** across web and mobile with strict response-time SLAs.

Led a **zero-downtime migration** of 100% of production infrastructure from Rackspace to **AWS**, cutting monthly hosting costs by **30%** and unlocking autoscaling.

Restructured a **100K+ line legacy codebase** into clean MVC; led and mentored **four developers** through code review, pairing, and release planning.

› Certifications

- Building with the Claude API · Claude Code in Action — Anthropic, 2026
- Go Design Patterns · Learning Go — LinkedIn Learning, 2024
- Go: The Complete Developer's Guide — Udemy, 2021
- Master RAG: Retrieval-Augmented Generation Systems — Udemy, 2025
- GraphQL with React: The Complete Developer's Guide — Udemy, 2023

› Writing

Blog — surajadsul.me · Essays on AI prompting, Claude Code, agent architecture, RAG systems, and developer productivity.